

UNDERSTANDING HACKING

An educational overview of cybersecurity concepts and hacker classifications

WHAT IS HACKING?

- Hacking is the practice of identifying and exploiting system vulnerabilities
- It can be malicious, ethical, or exploratory in nature
- Modern hacking targets networks, applications, and human psychology
- Security professionals study hacking to build better defenses



Hacking is a broad discipline covering both attacks and defense strategies.

TYPES OF HACKERS

- Black hat hackers act illegally for personal gain or damage
- White hat hackers test systems legally with permission
- Gray hat hackers operate in legal gray areas without clear authorization
- Script kiddies use existing tools without deep technical knowledge



COMMON ATTACK VECTORS

- Phishing exploits human trust through deceptive communications
- Malware includes viruses, ransomware, and spyware
- SQL injection targets database vulnerabilities in web applications
- Denial-of-service attacks overwhelm systems with traffic



DEFENSIVE SECURITY PRACTICES

UNDERSTANDING HACKING

- Regular patching closes known software vulnerabilities
- Multi-factor authentication prevents credential theft
- Network segmentation limits lateral movement after breach
- Security awareness training reduces social engineering success



Layered defenses address technical and human attack surfaces.

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LEGAL AND ETHICAL FRAMEWORKS

- Computer Fraud and Abuse Act criminalizes unauthorized access in the US
- GDPR and similar laws mandate data protection and breach notification
- Bug bounty programs provide legal channels for vulnerability disclosure
- Certifications like CEH and OSCP validate ethical hacking skills



KEY TAKEAWAYS

- Hacking spans a spectrum from criminal activity to essential security work
- Understanding attack methods enables effective defense strategies
- Legal authorization distinguishes ethical hacking from cybercrime
- Continuous learning is required as threats and technologies evolve



THANK YOU

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